



Sri Lanka Institute of Information Technology

PROJECT REGISTRATION FORM

The purpose of this form is to allow final year students of the B.Sc. (Hon) degree program to enlist in the final year project group. Enlisting in a project entail specifying the project title and the details of four members in the group, the internal supervisor (compulsory), external supervisor (may be from the industry) and indicating a brief description of the project. The description of the project entered on this form will not be considered as the formal project proposal. It should however indicate the scope of the project and provide the main potential outcome.

PROJECT TITLE

(As per accepted topic assessment form)

A smart mushroom incubation system for enhanced yield and sustainability in smallscale farming

RESEARCH GROUP

(As per accepted topic assessment form)

Machine Learning and Soft Computing

PROJECT NUMBER

25-26J-264

PROJECT GROUP MEMBER DETAILS:

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SUPERVISOR Details

Mr. Uditha Dharmakeerthi		21-06-2025
Name	Signature	Date

CO-SUPERVISOR Details

Mr. Hanojan Rajahrajasingh		21-06-2025
Name	Signature	Date

EXTERNAL SUPERVISOR Details

Name	Affiliation	Contact Address	Contact Numbers	Signature/Date

ACCEPTANCE BY CDAP MEMBER

Name	Signature	Date

PROJECT DETAILS

Brief Description of Your Research Problem

Mushroom cultivation has become an increasingly important agricultural activity due to its high nutritional value, medicinal benefits, and economic potential. However, successful mushroom production is highly dependent on maintaining precise environmental conditions such as temperature, humidity, carbon dioxide (CO₂) levels, light intensity, and air circulation. In traditional mushroom farming practices, especially in small- and medium-scale operations, these environmental parameters are monitored and controlled manually. This leads to inconsistencies, human errors, and an inability to maintain optimal growing conditions continuously.

One of the major challenges faced by mushroom farmers is the lack of real-time monitoring and automated control systems. Environmental conditions can fluctuate rapidly due to external factors such as weather changes, which directly affect the growth rate, quality, and yield of mushrooms. Improper temperature or humidity levels can lead to slow growth, contamination, or even complete crop failure. Additionally, high CO₂ levels and poor ventilation can negatively impact mushroom development, resulting in deformed or low-quality produce.

Another significant issue is the lack of technological adoption in rural farming communities. Many farmers rely on traditional knowledge and manual observation rather than data-driven approaches. This not only limits productivity but also increases labor effort and operational costs. Furthermore, the absence of proper monitoring systems makes it difficult for farmers to detect problems at an early stage, leading to delayed responses and reduced efficiency.

In addition to environmental control challenges, there is also a lack of centralized systems that provide insights, alerts, and recommendations to farmers. Without proper data analysis, farmers are unable to make informed decisions regarding optimal growing conditions, resource usage, and risk management.

Therefore, there is a strong need for an intelligent and automated system that can continuously monitor environmental parameters, provide real-time feedback, and maintain optimal conditions for mushroom growth. Integrating modern technologies such as IoT (Internet of Things), sensors, and smart control systems can significantly improve productivity, reduce manual effort, and ensure consistent quality in mushroom cultivation.

This research aims to address these challenges by developing a smart mushroom incubation system that leverages technology to enhance efficiency, sustainability, and overall crop yield.

Main expected outcomes of the project

The expected outcome of this project is the development of a smart and automated mushroom incubation system that ensures optimal environmental conditions for mushroom cultivation through continuous monitoring and intelligent control mechanisms.

The system will utilize IoT-based sensors to measure critical environmental parameters such as temperature, humidity, CO₂ levels, and light intensity in real time. These data will be processed and analyzed to maintain ideal growth conditions automatically using actuators such as fans, humidifiers, heaters, and ventilation systems. This automation will significantly reduce the need for manual intervention and minimize human errors.

Another key outcome is the development of a user-friendly application or dashboard that allows farmers to monitor environmental conditions remotely. The system will provide real-time updates, alerts, and notifications whenever conditions deviate from the optimal range. This enables farmers to take immediate action and prevent potential crop damage.

The project also aims to improve the overall yield and quality of mushroom production by maintaining stable and controlled conditions throughout the growth cycle. By ensuring consistency, the system helps reduce contamination risks and enhances productivity.

Additionally, the system will contribute to sustainable agricultural practices by optimizing resource usage such as water and energy. Farmers will be able to make data-driven decisions based on historical and real-time data insights provided by the system.

Ultimately, this solution aims to modernize traditional mushroom farming by introducing smart technology, making it more efficient, scalable, and economically beneficial for farmers.

WORKLOAD ALLOCATION

MEMBER 1

Vidanaraachchi M L

This component is responsible for developing an IoT-based environmental monitoring system combined with a Machine Learning model to predict the optimal fruiting stage of oyster mushrooms. The member will design and implement sensor integration to continuously capture environmental parameters such as temperature, humidity, CO₂ levels, and light intensity. The collected data will be pre-processed and used to train predictive models that determine fruiting readiness periods.

Additionally, this component includes developing mechanisms to generate real-time alerts and insights, which are delivered to farmers through a mobile application. The objective is to support informed decision-making, improve yield consistency, and optimize resource utilization by maintaining ideal environmental conditions.

MEMBER 2**Wijesinghe G A I W**

This component focuses on developing an IoT-based smart pest detection and automated response system. The member will implement a moving camera platform that continuously monitors mushroom racks and captures images for analysis.

These images will be processed using Convolutional Neural Network (CNN) models to detect pest presence in real time. Upon detection, the system will automatically trigger targeted pest control actions, such as localized neem spraying, to eliminate pests while minimizing chemical usage.

In addition, this component includes integrating a mobile application to provide instant alerts, pest location tracking, and farm insights. The goal is to enable precise, timely interventions that protect crop quality, reduce losses, and promote sustainable farming practices.

MEMBER 3**Waduge S W O P K**

This component focuses on developing a predictive advisory system to manage substrate conditions effectively. The member is responsible for implementing IoT sensors to monitor key parameters such as substrate moisture levels, pH values, and surrounding environmental factors in real time.

The collected data will be analyzed using advanced time-series forecasting models such as LSTM and ARIMA to predict future substrate conditions (12–24 hours ahead). Based on these predictions, an intelligent advisory engine will generate actionable recommendations along with confidence levels.

Furthermore, this component includes implementing a multi-channel alert system (dashboard, SMS, and mobile notifications) to ensure timely communication with farmers, enabling proactive decision-making and improved cultivation outcomes.

MEMBER 4**Peiris B H A S**

This component is responsible for designing and implementing an AI-based vision system to detect fungal contamination in mushroom cultivation. The member will develop a motorized camera mechanism that performs systematic scanning of mushroom racks and captures images of individual mushroom packets.

These images will be processed using image processing techniques and analyzed using a Convolutional Neural Network (CNN) model trained to identify contamination types such as green mold, cobweb disease, and discoloration.

The system will provide accurate detection results and notify farmers in real time, enabling early intervention to prevent the spread of contamination and minimize crop losses.

DECLARATION

"We declare that this project is our own original work and has been carried out by the group members. This project does not contain any material previously submitted for a degree or diploma in any other university or institution. All sources of information used in this research have been properly acknowledged. We understand that any form of plagiarism or unauthorized reproduction will be treated as an offense under institutional regulations and may result in disciplinary action."

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